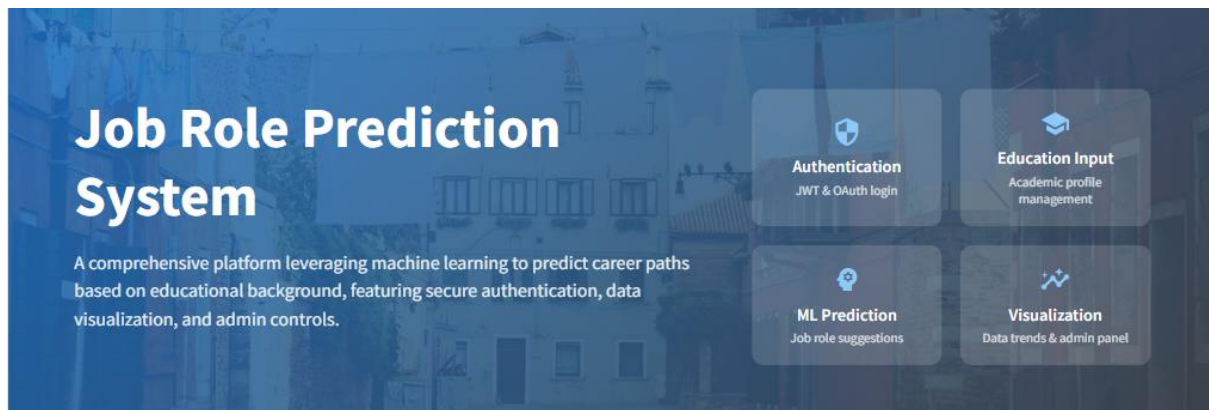


1.Title: Predicting Suitable Job Roles from Education Details



2. Project Statement:

In today's dynamic job market, students and professionals often struggle to identify career paths that align with their educational background and skill set. This project aims to bridge the gap between education and employment by analyzing a user's academic history and suggesting suitable job roles. Leveraging machine learning and data analytics, the system will predict job roles that best match the user's qualifications, enabling informed career planning and better utilization of one's skill set.

3. Outcomes:

- Predictive model that maps education history to job roles.
- User-friendly interface to input education details and get career recommendations.
- Admin dashboard to manage datasets, model training, and user queries.
- Visualization of the correlation between education fields and career outcomes.
- Secure role-based user access and history tracking.

4. Modules to be Implemented:

1. User Authentication and Profile Management

- User registration and login using JWT and Google OAuth.
- Role-based access for Users and Admins.
- User profile to store education history and job prediction history.

2. Education Details Input & Preprocessing Module

- Input forms for users to enter their academic data (degree, specialization, CGPA, certifications).

- Data validation and preprocessing pipeline (encoding, normalization) before feeding into the ML model.

3. Job Role Prediction Module

- ML model to predict suitable job roles using user input.
- Backend logic to query and return top 3-5 job matches with confidence scores.
- Option to give alternative job role suggestions and reasoning (skills gap, overlapping domains).

4. Visualization and Career Insights

- Graphs showing education-to-job trends (e.g., which degrees lead to which roles).
- Pie charts or bar graphs of job domains per education field.
- Compare user data with similar profiles to show potential pathways.

5. Admin Dashboard & Model Management

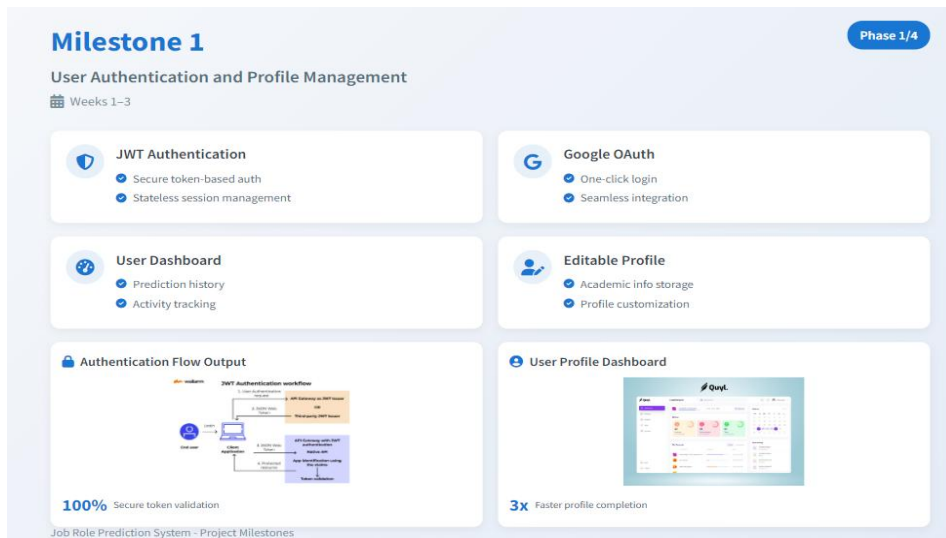
- Admin can upload new training data and retrain the model.
- View logs of user queries and predictions.
- Manage flagged or incorrect predictions for review.

5. Week-wise Module Implementation and High-Level Requirements with sample screenshots :

Milestone 1: Weeks 1–3

Module 1: User Authentication and Profile Management

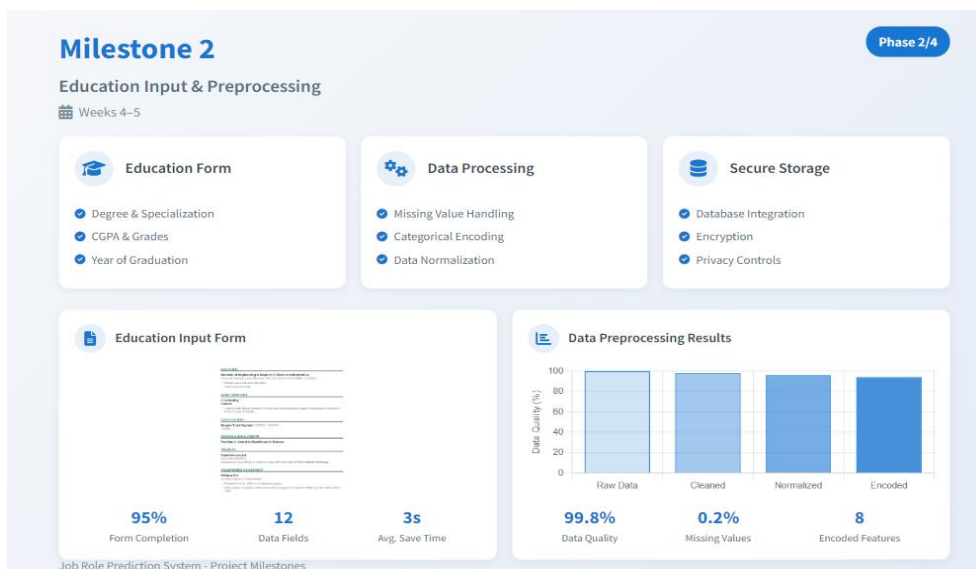
- JWT-based secure user login and registration.
- Google OAuth login integration.
- Dashboard with user history of predictions.
- Editable profile storing academic info.



Milestone 2: Weeks 4–5

Module 2: Education Input & Preprocessing

- Form to capture education details (degree, specialization, CGPA, year of graduation, etc.).
- Input validation and preprocessing logic (e.g., handling missing values, encoding categorical fields).
- Store data securely in user profile.

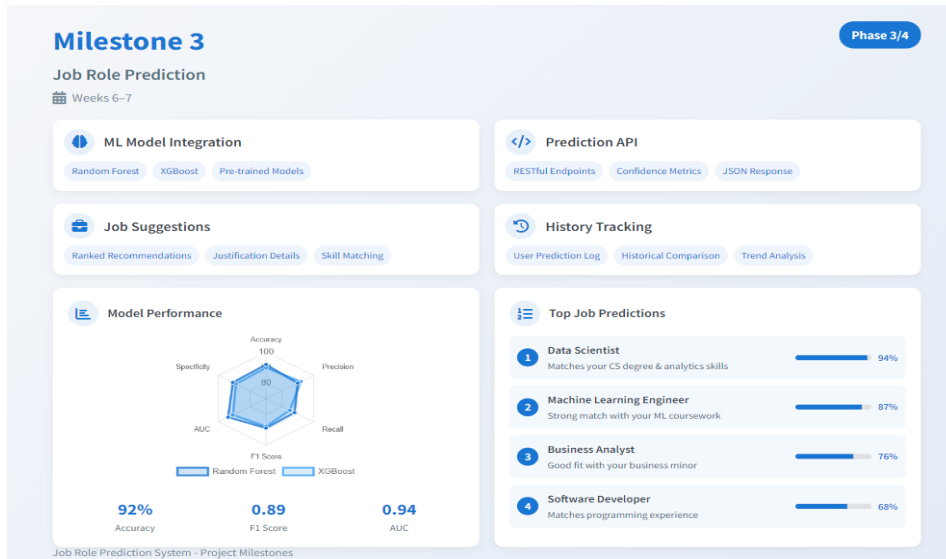


Milestone 3: Weeks 6–7

Module 3: Job Role Prediction

- Integrate trained ML model (e.g., Random Forest, XGBoost).

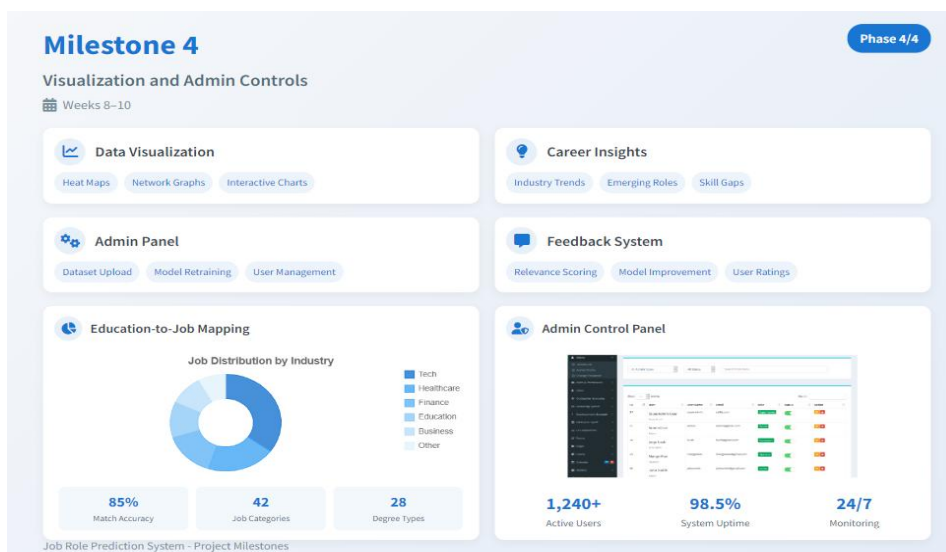
- Backend API to return predictions with confidence scores.
- Display top job suggestions with explanation/justification.
- Save result to user history.



Milestone 4: Weeks 8-10

Module 4: Visualization and Admin Controls

- Data visualizations of common education-to-job mappings.
- Insights on career transitions and trends based on dataset.
- Admin panel: upload dataset, trigger retraining, manage users and predictions.
- Feedback system for users to rate suggestion relevance.



6. Evaluation Criteria:

Milestone 1 Evaluation (Week 3):

- User registration, login, and profile creation fully functional.
- Secure JWT and Google OAuth implemented.
- User data is stored and retrievable.

Milestone 2 Evaluation (Week 5):

- Education input system functional with robust data validation.
- Clean and usable interface for data entry.
- Inputs are properly processed and stored.

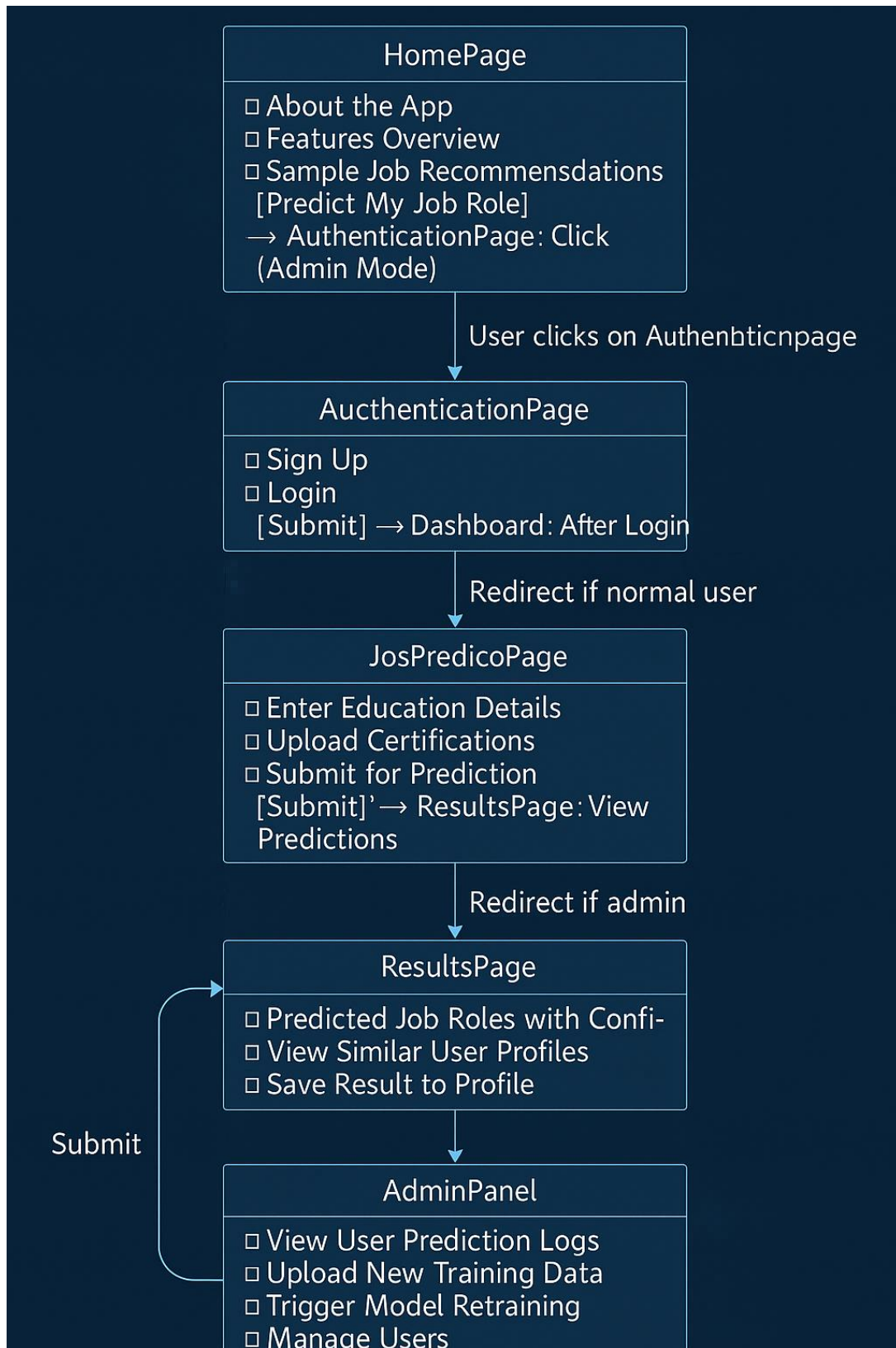
Milestone 3 Evaluation (Week 7):

- Model predictions are integrated and working with real input data.
- Output shows top roles with confidence and justification.
- Prediction history is saved and accessible.

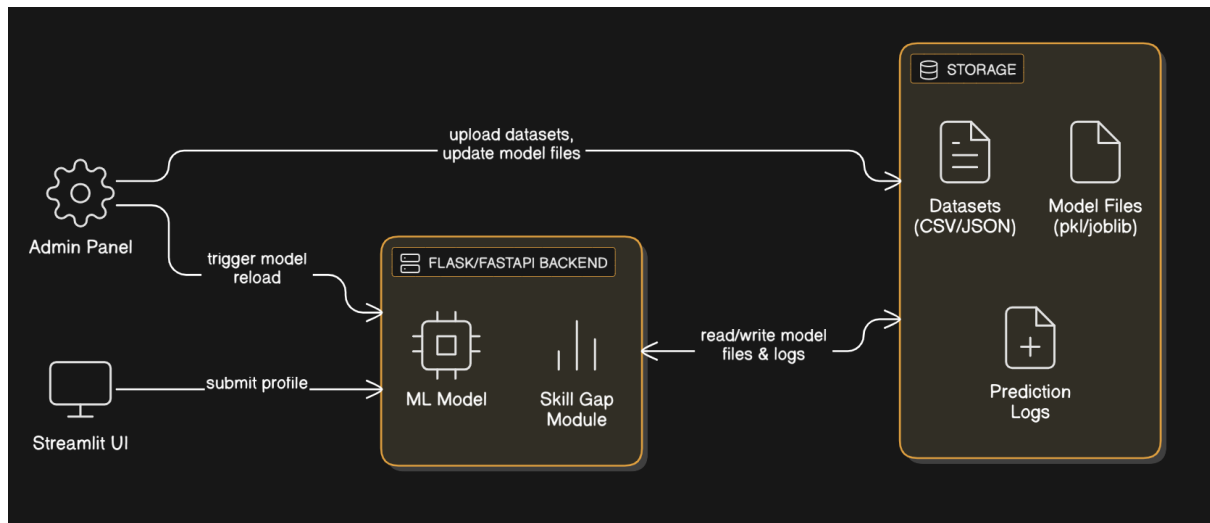
Milestone 4 Evaluation (Week 10):

- Data visualizations and trends are accurately displayed.
 - Admin can manage models, data, and user activity.
 - Feedback and improvements based on user interactions are possible.
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7. Workflow Diagram



8. Architecture Diagram



9. Database Schema

